Hamiltonian-Inspired Energy Dissipation and Chunk-Wise Variational State Coupling for Efficient Multimodal State-Space Models

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***Abstract*—State-Space Models (SSMs), including structured state spaces and State-Space Duality (SSD) architectures, have emerged as computationally efficient sub-quadratic alternatives to quadratic multi-head self-attention. Despite their theoretical appeal in unimodal sequence processing, deploying state-space recurrence within cross-modal vision-language metric learning introduces two practical challenges: (1) sensitivity to representation perturbations and background token noise across sequence representations, and (2) potential misalignment between modality-specific boundary representations, which may affect downstream retrieval alignment. In this work, we propose a multimodal state-space framework comprising two core components: (i) the Hamiltonian-Inspired Energy Dissipation Operator (HEDO), a learned discrete coordinate-momentum transformation with positive damping ($\beta = 0.05, \Delta t = 0.1, \gamma = 0.1$) that introduces a Hamiltonian-inspired dissipative inductive bias, and (ii) Chunk-Wise Variational State Coupling (HVSC), which performs mask-weighted aggregation of chunk boundary states ($C = 16, d_{\text{state}} = 64$) and regularizes modality-specific Gaussian latent distributions through symmetric Kullback-Leibler (KL) divergence during training while preserving strictly independent, deterministic unimodal inference at test time. We conduct a controlled 12-run factorial multi-seed benchmark (4 model configurations $\times$ 3 seeds $: 42, 43, 44$) on the standard Flickr8k dataset with frozen ViT-B/16 and RoBERTa-base backbones. The full HEDO-HVSC architecture achieves a test Mean Recall of $48.28\% \pm 0.59\%$ ($26.73\% \pm 1.87\%$ I2T R@1, $56.83\% \pm 0.47\%$ I2T R@5, $70.47\% \pm 0.80\%$ I2T R@10, $20.71\% \pm 0.18\%$ T2I R@1, $49.97\% \pm 0.40\%$ T2I R@5, $64.96\% \pm 0.34\%$ T2I R@10), demonstrating retrieval performance comparable to the Custom SSD-Style Baseline ($48.25\% \pm 0.47\%$, mean paired delta $+0.03\% \pm 0.20\%$). In corruption stress testing, the full model exhibits a mixed robustness profile, outperforming the baseline by $+0.50\%$ under severe Gaussian noise ($\sigma = 0.10$). Furthermore, empirical latency profiling across sequence lengths $L \in [16, 256]$ shows approximately linear scaling, consistent with the algorithmically linear sequential recurrence.**

***Index Terms*—Multimodal retrieval, image-text retrieval, state-space models, state-space duality, Hamiltonian-inspired dynamics, variational representation learning, vision-language alignment.**

I. Introduction

CROSS-MODAL image-text retrieval represents a foundational problem in multimodal machine learning, serving as the computational backbone of modern image search engines, visual question answering, cross-modal digital asset management, and autonomous embodied agents [1]–[3]. The primary goal is to project high-dimensional visual patch sequences and textual token sequences into a joint metric embedding space where semantic alignment corresponds directly to vector proximity [4], [5].

A. Computational Challenges in Multimodal Transformers

Over the past decade, Transformer architectures based on multi-head self-attention and cross-attention have served as the standard foundation for vision-language models [6]–[8]. While self-attention provides high representational capacity by computing pairwise affinities across all token positions, its computational complexity and memory footprint scale quadratically, $\mathcal{O}(L^2)$, with sequence length L [9]. In multimodal applications where high-resolution visual inputs are partitioned into hundreds of spatial tokens and textual captions extend across multiple sentences, this quadratic growth creates substantial computational overhead during training and deployment [10], [11].

B. State-Space Models as Sub-Quadratic Alternatives

To mitigate quadratic scaling, linear-time sequence models—such as Structured State Spaces (S4) [12], S5 [13], Mamba [14], and State-Space Duality (SSD) [15]—have been introduced. SSM formulations can provide linear algorithmic complexity $\mathcal{O}(L)$ with respect to sequence length, with implementation-dependent training and inference strategies, contrasting with quadratic attention [13], [16].

C. Challenges in Multimodal State-Space Recurrence

Despite their efficiency in unimodal sequence modeling, applying state-space recurrence to multimodal metric learning introduces specific challenges:

1) **Representation Perturbation Sensitivity:** Standard discrete recurrent state transitions propagate token variations across the sequence axis. In visual patch sequences, localized sensor noise, background clutter, and photometric variations can propagate across recurrent steps, leading to trajectory drift in hidden state representations [17].
2) **Potential Boundary Representation Misalignment:** Unidirectional recurrence across separate visual and

textual encoders may accumulate modality-specific representational drift. In textual streams, variable sentence lengths require zero-padding; unweighted sequence-end pooling allows uninformative pad tokens to influence final representations, potentially degrading retrieval alignment [4], [18].

D. Contributions of This Work

To investigate these challenges, this paper presents HEDO-HVSC, a multimodal state-space framework combining dissipative coordinate-momentum updates with variational chunk boundary coupling. The primary contributions of this work are:

1) **Hamiltonian-Inspired Energy Dissipation Operator (HEDO):** We formulate a learned discrete coordinate-momentum transformation parameterized by weight matrices $\mathbf{W}_q, \mathbf{W}_p$, integration step $\Delta t = 0.1$, damping coefficient $\beta = 0.05$, and perturbation scale $\gamma = 0.1$. HEDO introduces an explicit damping term intended to regulate momentum magnitude prior to sequence recurrence.
2) **State-Continuous Chunk-Wise SSD-Style Recurrence:** We design a custom PyTorch recurrent state-space block ($C = 16, d_{\text{state}} = 64$) inspired by the state-space recurrence perspective of SSD that maintains hidden-state continuity across chunk boundaries, avoiding state resets between consecutive chunks.
3) **Chunk-Wise Variational State Coupling (HVSC):** We introduce mask-weighted boundary-state aggregation to reduce padding-induced aggregation bias, parameterizing Gaussian latent distributions regularized via symmetric KL divergence in FP32 precision during training, while ensuring strictly unimodal, deterministic inference at test time.
4) **Controlled Multi-Seed Factorial Benchmark:** We execute a 12-run controlled factorial benchmark (4 model configurations $\times$ 3 random seeds : $42, 43, 44$) on the standard Flickr8k dataset [19] with frozen ViT-B/16 and RoBERTa-base backbones, accompanied by secondary corruption testing and empirical latency scaling analysis.

II. Literature Survey

A. Vision-Language Metric Learning

Cross-modal representation learning aims to bridge the semantic gap between visual perception and natural language. Foundational methods, including VSE++ [4] and SCAN [5], established metric learning frameworks utilizing hard negative ranking losses and stacked cross-attention networks.

The introduction of contrastive vision-language pretraining by CLIP [1] and ALIGN [3] demonstrated that training dual-encoder architectures on web-scale image-caption pairs using symmetric InfoNCE loss yields robust representations across diverse downstream tasks. Subsequent works, such as BLIP [2] and BLIP-2 [10], combined contrastive objectives with multimodal cross-attention decoders to enhance fine-grained visual reasoning. However, standard cross-attention architectures incur quadratic complexity $\mathcal{O}(L^2)$ with respect to sequence length, creating computational bottlenecks when scaling to high-resolution image patches and long textual documents.

B. Structured State Space Models

To circumvent the quadratic complexity of self-attention, State-Space Models (SSMs) have been introduced for deep sequence modeling. The Structured State Space Sequence model (S4) [12] formalized continuous-time linear time-invariant (LTI) systems using HiPPO matrix initializations [12], achieving strong performance on long-range benchmarks. S5 [13] simplified S4 by utilizing parallel prefix scans over diagonalized state matrices.

Mamba [14] introduced selective state spaces, allowing the recurrent matrices $\mathbf{B}, \mathbf{C}$, and decay step Δ to be dynamic functions of the input tokens. More recently, State-Space Duality (SSD) and Mamba-2 [15] established the mathematical equivalence between 1-D structured masked attention and continuous recurrence. In this paper, we extend state-space recurrence to multimodal metric learning using a custom PyTorch SSD-style recurrent formulation inspired by the state-space recurrence perspective of SSD, rather than native Mamba-2/SSD CUDA kernels.

C. Hamiltonian Mechanics and Neural Dynamical Systems

Hamiltonian Neural Networks (HNNs) [20] and Lagrangian Neural Networks [21] embed fundamental energy-conservation principles into deep learning architectures by enforcing symplectic gradient updates. Port-Hamiltonian systems [22] incorporate positive semi-definite damping matrices $\mathbf{D} \succeq 0$ to model dissipative continuous neural dynamical systems. In our architecture, HEDO adapts discrete dissipative mechanics into a parameter-efficient sequence transformation operator to introduce an explicit dissipative inductive bias for sequence representations.

D. Variational Representation Learning

Variational Autoencoders (VAEs) [23] and the Variational Information Bottleneck (VIB) [24] formalize latent representation learning by optimizing trade-offs between mutual information and compression. In this work, HVSC parameterizes Gaussian latent distributions from recurrent chunk boundary states, using symmetric KL divergence to align cross-modal boundary representations during training.

E. Identified Research Gap

While existing paradigms have advanced individual domains, their intersection exposes critical limitations:

1) **Quadratic Overhead vs. Unimodal SSMs:** Modern multimodal Transformers incur quadratic attention cost $\mathcal{O}(L^2)$, whereas SSMs have predominantly been investigated for unimodal sequence processing without addressing multimodal cross-modal metric alignment.

2) **Continuous vs. Discrete Dissipation:** Prior Hamiltonian neural models focus on continuous differential equations; their discrete formulation as parameter-efficient pre-filters for sequence models remains largely unexplored.
3) **Boundary Alignment:** Variational alignment techniques have not been formulated to regularize chunk-boundary hidden states in state-continuous recurrent architectures.

Therefore, this work investigates whether a Hamiltonian-inspired dissipative transformation (HEDO) combined with chunk-wise variational state coupling (HVSC) can provide an effective, parameter-efficient multimodal state-space retrieval framework.

III. Problem Formulation and Theoretical Background

A. Cross-Modal Image-Text Metric Learning

Let $\mathcal{D} = \{(\mathbf{I}_i, \mathcal{T}_i)\}_{i=1}^{N}$ denote a multimodal dataset where each image $\mathbf{I}_i \in \mathbb{R}^{3\times H\times W}$ is annotated with a set of M ground-truth descriptive captions $\mathcal{T}_i = \{\mathbf{T}_{i,m}\}_{m=1}^{M}$. The metric learning objective is to optimize parameterized visual and textual encoders, $f_\theta(\mathbf{I}) \in \mathbb{S}^{d-1}$ and $g_\phi(\mathbf{T}) \in \mathbb{S}^{d-1}$, mapping inputs onto the unit hypersphere such that semantic relevance correlates with cosine similarity:

$$s(\mathbf{I}, \mathbf{T}) = f_\theta(\mathbf{I})^\top g_\phi(\mathbf{T}) = \cos\angle\left(f_\theta(\mathbf{I}), g_\phi(\mathbf{T})\right) \tag{1}$$

B. State-Space Duality and Sequence Recurrence

Continuous-time state-space models map a 1-D continuous input signal $u(t) \in \mathbb{R}$ to an output signal $y(t) \in \mathbb{R}$ via a latent state $\mathbf{h}(t) \in \mathbb{R}^N$:

$$\dot{\mathbf{h}}(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}u(t) \tag{2}$$

$$y(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}u(t) \tag{3}$$

Discretizing this continuous system using Zero-Order Hold (ZOH) with step size Δ yields the discrete recurrent equations:

$$\mathbf{h}_t = \overline{\mathbf{A}}\mathbf{h}_{t-1} + \overline{\mathbf{B}}u_t \tag{4}$$

$$y_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}u_t \tag{5}$$

where $\overline{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\overline{\mathbf{B}} = \int_0^\Delta \exp((\Delta-\tau)\mathbf{A})\mathbf{B}\,d\tau$. State-Space Duality (SSD) [15] proves that when $\mathbf{A}$ is structured as a scalar-times-identity matrix, discrete recurrence can be computed via structured block matrix multiplication, combining parallel training scans with linear-time inference. The present benchmark implementation uses an explicit sequential PyTorch recurrence; optimized parallel SSD kernels are outside the scope of this implementation.

C. Hamiltonian and Dissipative Mechanics

In classical mechanics, a physical system with generalized coordinates $\mathbf{q} \in \mathbb{R}^d$ and conjugate momenta $\mathbf{p} \in \mathbb{R}^d$ is governed by the scalar Hamiltonian function $\mathcal{H}(\mathbf{q}, \mathbf{p}) = \mathcal{T}(\mathbf{p}) + \mathcal{V}(\mathbf{q})$. For an open dissipative system with damping matrix $\mathbf{D} \succ 0$, the state dynamics evolve as:

$$\begin{bmatrix}\dot{\mathbf{q}}\\ \dot{\mathbf{p}}\end{bmatrix} = \begin{bmatrix}\mathbf{0} & \mathbf{I}\\ -\mathbf{I} & -\mathbf{D}\end{bmatrix}\begin{bmatrix}\nabla_{\mathbf{q}}\mathcal{H}\\ \nabla_{\mathbf{p}}\mathcal{H}\end{bmatrix} \tag{6}$$

The total energy derivative evaluates to $\dot{\mathcal{H}} = -\nabla_{\mathbf{p}}\mathcal{H}^\top\mathbf{D}\nabla_{\mathbf{p}}\mathcal{H} \le 0$, ensuring continuous energy dissipation. This continuous-time dissipation property motivates our construction but does not directly imply monotonic energy decay for the proposed discrete learned HEDO update. In our architecture, we adapt these continuous physical principles into a discrete learnable sequence transformation operator. The term "energy dissipation" refers to the damping mechanism and physical motivation, not to a guaranteed monotonic decrease of the discrete empirical energy.

IV. Proposed HEDO-HVSC Architecture

A. Architectural Pipeline

The overall architecture of HEDO-HVSC (Fig. 1) processes paired images $\mathbf{I} \in \mathbb{R}^{3\times 224\times 224}$ and textual captions $\mathbf{T}$ through four coordinated stages:

1) **Frozen Backbone Feature Extraction:** Input images and tokenized text are mapped into continuous sequence representations using frozen pretrained encoders (ViT-B/16 and RoBERTa-base).
2) **Dissipative Hamiltonian Transformation (HEDO):** Token representations are processed through coordinate-momentum dissipative updates intended to reduce sensitivity to representation perturbations.
3) **State-Continuous Chunk-Wise Recurrence (SSD):** Sequences are partitioned into non-overlapping chunks of size $C = 16$ and processed with inter-chunk hidden-state continuity.
4) **Variational Chunk Boundary State Coupling (HVSC):** Valid chunk boundary states are aggregated with attention-mask weighting and used to parameterize modality-specific Gaussian latent distributions.

B. Frozen Pretrained Feature Extraction

Given an input image $\mathbf{I} \in \mathbb{R}^{3\times 224\times 224}$, a frozen Vision Transformer (ViT-B/16) extracts patch tokens with a classification token, projected to $d_{\text{model}} = 128$:

$$\mathbf{X}_I^{(0)} = \text{LayerNorm}\left(\mathbf{W}_{\text{proj},I}\,\text{ViT}(\mathbf{I})\right) \in \mathbb{R}^{L_I\times d_{\text{model}}} \tag{7}$$

where $L_I = 197$. For a paired caption $\mathbf{T}$, a frozen RoBERTa-base encoder extracts contextual representations:

$$\mathbf{X}_T^{(0)} = \text{LayerNorm}\left(\mathbf{W}_{\text{proj},T}\,\text{RoBERTa}(\mathbf{T})\right) \in \mathbb{R}^{L_T\times d_{\text{model}}} \tag{8}$$

where $L_T = 64$. A binary padding mask $\mathbf{M}_T \in \{0,1\}^{B\times L_T}$ identifies valid token positions (1 for valid tokens, 0 for pad tokens).

C. Hamiltonian-Inspired Energy Dissipation Operator (HEDO)

To regulate representation perturbations prior to sequence recurrence, each token representation $\mathbf{x} \in \mathbb{R}^{d_{\text{model}}}$ initializes generalized coordinates $\mathbf{q}_0 \in \mathbb{R}^{d_{\text{model}}}$ and conjugate momenta $\mathbf{p}_0 \in \mathbb{R}^{d_{\text{model}}}$:

$$\mathbf{q}_0 = \mathbf{x}, \quad \mathbf{p}_0 = \tanh(\mathbf{W}_p\mathbf{q}_0 + \mathbf{b}_p) \tag{9}$$

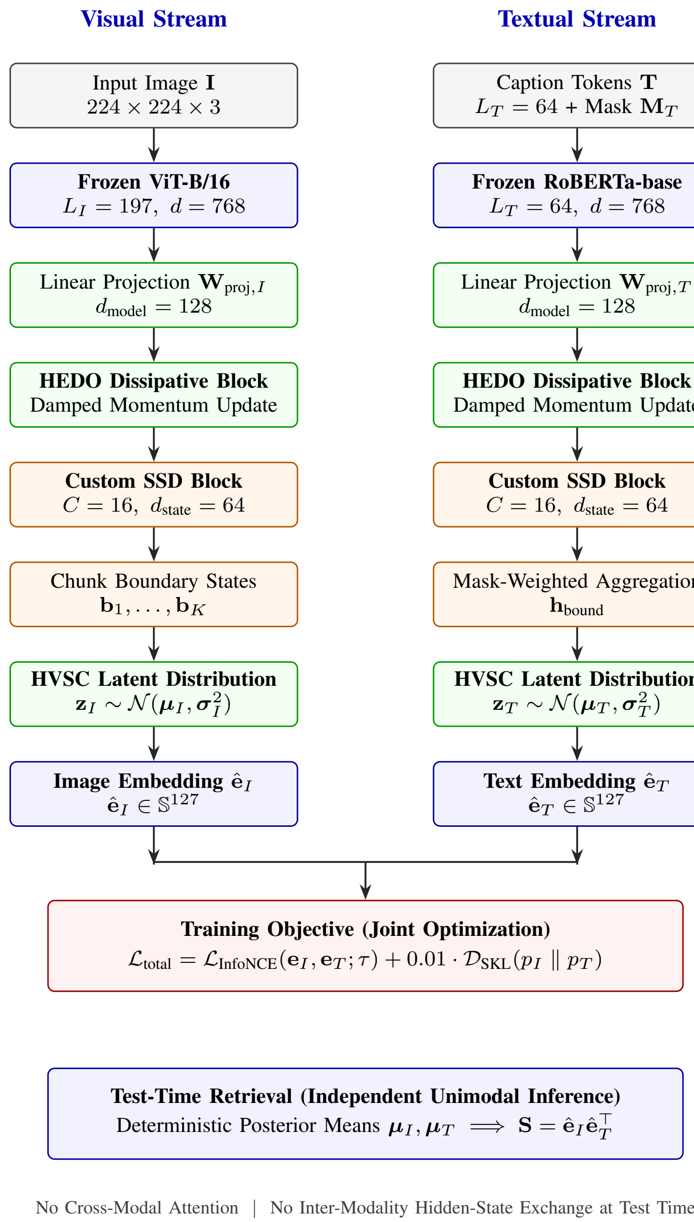

Fig. 1. Overall architectural pipeline of the proposed HEDO-HVSC multimodal state-space framework. Visual patch sequences ($L_I = 197$) and textual token sequences ($L_T = 64$) are extracted from frozen ViT-B/16 and RoBERTa-base encoders, projected to $d_{\text{model}} = 128$, stabilized via HEDO coordinate-momentum dissipation, and processed through custom chunk-wise SSD recurrence blocks ($C = 16, d_{\text{state}} = 64$). During training, mask-weighted boundary states parameterize Gaussian latent distributions regularized via symmetric KL divergence $\mathcal{D}_{\text{SKL}}$ alongside multi-positive InfoNCE contrastive alignment. At test time, retrieval evaluates deterministic posterior means strictly unimodally with zero cross-modal communication.

where $\mathbf{W}_p \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ is a learned projection matrix. We define quadratic potential $\mathcal{V}(\mathbf{q}) = \frac{1}{2}\|\mathbf{q}\|_2^2$ and kinetic energy $\mathcal{T}(\mathbf{p}) = \frac{1}{2}\|\mathbf{p}\|_2^2$, with conservative potential gradient field parameterized via $\nabla_{\mathbf{q}}\mathcal{V} = \tanh(\mathbf{W}_q\mathbf{q} + \mathbf{b}_q)$.

The discrete dissipative update integrates coordinate-momentum dynamics over step size $\Delta t = 0.1$ with damping coefficient $\beta = 0.05$:

$$\mathbf{p}_{k+1} = (1 - \beta\Delta t)\,\mathbf{p}_k - \Delta t \tanh(\mathbf{W}_q\mathbf{q}_k + \mathbf{b}_q) \tag{10}$$

$$\mathbf{q}_{k+1} = \mathbf{q}_k + \Delta t\,\mathbf{p}_{k+1} \tag{11}$$

The dissipated sequence representation is reconstructed with residual coupling:

$$\mathbf{X}^{(1)} = \mathbf{X}^{(0)} + \gamma\,\text{LayerNorm}(\mathbf{q}_K) \tag{12}$$

where $\gamma = 0.1$ and $K = 1$ step.

Empirical Diagnostic Trajectory: In pre-training diagnostic evaluation, unforced multi-step simulation on image patch representations yielded the trajectory $\mathcal{H} \in [170.824, 170.651, 170.927, 171.646, 172.801, 174.387]$, while text representations yielded $\mathcal{H} \in [5.849, 5.796, 5.774, 5.783, 5.823, 5.892]$. While HEDO changes representation energy substantially, the discrete trajectory is not strictly monotonically decreasing; HEDO is therefore presented as a parameterized coordinate-momentum transformation with a dissipative inductive bias, rather than a system with guaranteed monotonic energy decay.

D. State-Continuous Chunk-Wise SSD Recurrence

The sequence $\mathbf{X}^{(1)} \in \mathbb{R}^{L \times d_{\text{model}}}$ is divided into $K = \lceil L/C \rceil$ non-overlapping chunks of length $C = 16$. Within each chunk $k \in \{1, \ldots, K\}$, the recurrent state $\mathbf{h}_{k,t} \in \mathbb{R}^{d_{\text{state}}}$ evolves according to discrete-time recurrence ($t \in \{1, \ldots, C\}$):

$$\mathbf{h}_{k,t} = m_{k,t} \left(\mathbf{A}_{\text{decay}} \odot \mathbf{h}_{k,t-1} + \mathbf{B}_{k,t} \right) + (1 - m_{k,t}) \mathbf{h}_{k,t-1} \tag{13}$$

$$\mathbf{y}_{k,t} = \mathbf{h}_{k,t} \tag{14}$$

where $\mathbf{A}_{\text{decay}} = \exp(-\exp(\mathbf{A}_{\log})) \in (0,1)^{d_{\text{state}}}$, $\mathbf{B}_{k,t} = \mathbf{W}_B \mathbf{x}_{k,t} \in \mathbb{R}^{d_{\text{state}}}$, $d_{\text{state}} = 64$, and $m_{k,t} \in \{0,1\}$ denotes token validity (preserving state $\mathbf{h}_{k,t} = \mathbf{h}_{k,t-1}$ for zero-padded tokens).

Inter-Chunk Continuity: Unlike standard chunked models that reset hidden states between chunks, our formulation propagates boundary states:

$$\mathbf{h}_{k+1,0} = \mathbf{h}_{k,C} \tag{15}$$

with $\mathbf{h}_{1,0} = \mathbf{0}$. The final hidden state of chunk k is extracted as boundary state $\mathbf{b}_k = \mathbf{h}_{k,C} \in \mathbb{R}^{d_{\text{state}}}$.

E. Chunk-Wise Variational State Coupling (HVSC)

For the textual stream, variable caption lengths mean trailing chunks may contain padding tokens. To prevent padding corruption, we compute valid token counts per chunk:

$$n_k = \sum_{j=1}^{C} \mathbf{M}_{T,(k-1)C+j} \tag{16}$$

and assign normalized chunk weights:

$$w_k = \frac{n_k}{\sum_{j=1}^{K} n_j + \epsilon} \tag{17}$$

The mask-weighted boundary representation is:

$$\mathbf{h}_{\text{bound},T} = \sum_{k=1}^{K} w_k \mathbf{b}_{T,k} \in \mathbb{R}^{d_{\text{state}}} \tag{18}$$

For visual sequences where all $L_I = 197$ tokens are valid, uniform weighting $w_k = 1/K$ is applied.

The aggregated boundary representation parameterizes Gaussian latent distributions:

$$\boldsymbol{\mu} = \mathbf{W}_\mu \mathbf{h}_{\text{bound}} + \mathbf{b}_\mu \in \mathbb{R}^{d_{\text{latent}}} \tag{19}$$

$$\log \boldsymbol{\sigma}^2 = \text{Clamp}\left(\mathbf{W}_{\text{logvar}} \mathbf{h}_{\text{bound}} + \mathbf{b}_{\text{logvar}}, -10, 10 \right) \tag{20}$$

where $d_{\text{latent}} = 64$.

During training, latent vectors are sampled via the reparameterization trick:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{21}$$

The final embedding vectors $\mathbf{e}_I, \mathbf{e}_T \in \mathbb{S}^{d_{\text{model}}-1}$ combine mean-pooled recurrent representations with variational latent projections:

$$\mathbf{e}_I = \text{Normalize}\left(\text{LayerNorm}(\mathbf{h}_{\text{pool},I} + \alpha \mathbf{W}_{\text{out}} \mathbf{z}_I) \right) \tag{22}$$

$$\mathbf{e}_T = \text{Normalize}\left(\text{LayerNorm}(\mathbf{h}_{\text{pool},T} + \alpha \mathbf{W}_{\text{out}} \mathbf{z}_T) \right) \tag{23}$$

where $\alpha = 0.1$ and $\text{Normalize}(\mathbf{v}) = \mathbf{v}/\|\mathbf{v}\|_2$.

Symmetric KL Regularization: To align the visual and textual latent spaces during training, we compute symmetric Kullback-Leibler divergence in FP32 precision to prevent underflow:

$$\mathcal{D}_{\text{SKL}}(p_I \parallel p_T) = \frac{1}{2} \left[\mathcal{D}_{\text{KL}}(p_I \parallel p_T) + \mathcal{D}_{\text{KL}}(p_T \parallel p_I) \right] \tag{24}$$

where for diagonal Gaussians $p_I = \mathcal{N}(\boldsymbol{\mu}_I, \boldsymbol{\Sigma}_I)$ and $p_T = \mathcal{N}(\boldsymbol{\mu}_T, \boldsymbol{\Sigma}_T)$:

$$\mathcal{D}_{\text{KL}}(p_I \parallel p_T) = \frac{1}{2} \sum_{j=1}^{d_{\text{latent}}} \left[\log \frac{\sigma_{T,j}^2}{\sigma_{I,j}^2} + \frac{\sigma_{I,j}^2 + (\mu_{I,j} - \mu_{T,j})^2}{\sigma_{T,j}^2} - 1 \right] \tag{25}$$

Test-Time Deterministic Unimodal Inference: During test-time retrieval (Fig. 1), stochastic sampling is omitted, and the posterior mean vectors $\boldsymbol{\mu}_I, \boldsymbol{\mu}_T$ are evaluated deterministically:

$$\hat{\mathbf{e}}_I = \text{Normalize}\left(\text{LayerNorm}(\mathbf{h}_{\text{pool},I} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_I) \right) \tag{26}$$

$$\hat{\mathbf{e}}_T = \text{Normalize}\left(\text{LayerNorm}(\mathbf{h}_{\text{pool},T} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_T) \right) \tag{27}$$

Because visual and textual embeddings are computed completely independently without cross-modal attention or hidden-state exchange, test-time gallery retrieval is strictly unimodal.

F. Training Objective

The total loss combines a multi-positive InfoNCE contrastive objective with symmetric KL regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{InfoNCE}} + \lambda_{\text{KL}} \, \mathcal{D}_{\text{SKL}}(p_I \parallel p_T) \tag{28}$$

where $\lambda_{\text{KL}} = 0.01$. The multi-positive InfoNCE loss accounts for multiple captions per image (5 captions per image in Flickr8k):

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2B} \sum_{i=1}^{B} \left[\log \frac{\sum_{j \in \mathcal{P}(i)} \exp(s_{i,j} \cdot \alpha_{\text{scale}})}{\sum_{k=1}^{5B} \exp(s_{i,k} \cdot \alpha_{\text{scale}})} + \log \frac{\exp(s_{i,i} \cdot \alpha_{\text{scale}})}{\sum_{k=1}^{B} \exp(s_{k,i} \cdot \alpha_{\text{scale}})} \right] \tag{29}$$

where $s_{i,j} = \hat{\mathbf{e}}_{I,i}^\top \hat{\mathbf{e}}_{T,j}$ denotes cosine similarity, $\mathcal{P}(i)$ denotes the positive caption index set for image i, and the learnable logit scale parameter $\alpha_{\text{scale}} = \exp(\ell)$ is clamped to a maximum value of 100.0 ($\alpha_{\text{scale}} \leq 100.0$).

V. Experimental Setup and Dataset Protocol

A. Dataset Partition and Provenance Protocol

Experiments are conducted on the standard Flickr8k benchmark introduced by Hodosh et al. [19], comprising 8,000 photographic images paired with 5 human-annotated captions each (40,000 total captions). We utilize a predefined 6,000 / 1,000 / 1,000 train / validation / test partition protocol summarized in Table I and Fig. 2. Split integrity was strictly verified via image identifier hashing to guarantee zero cross-split overlap ($\text{Train} \cap \text{Val} = \emptyset$, $\text{Train} \cap \text{Test} = \emptyset$, $\text{Val} \cap \text{Test} = \emptyset$).

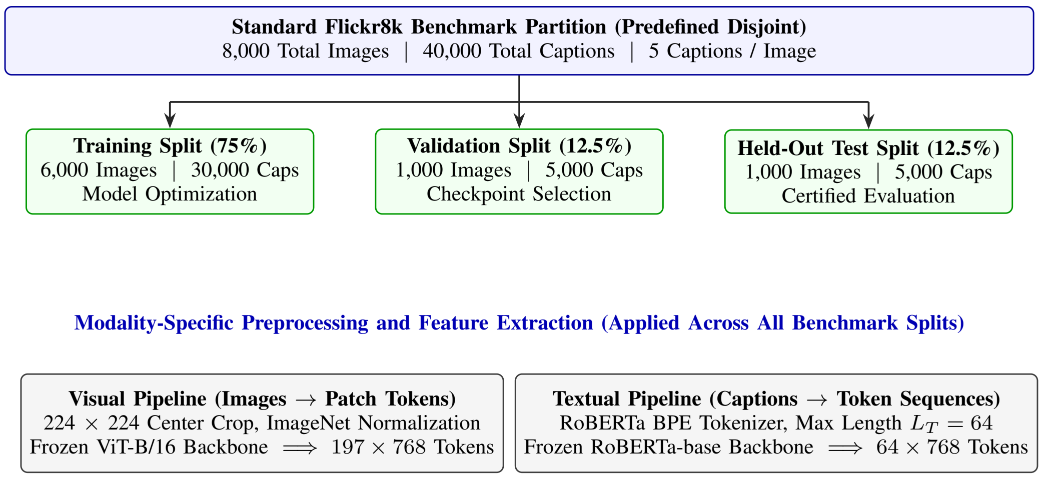

Fig. 2. Predefined Flickr8k dataset partition and modality-specific feature extraction pipeline. The 8,000 photographic images and 40,000 human-annotated captions are partitioned into verified disjoint training (6,000), validation (1,000), and held-out test (1,000) splits with zero cross-split leakage, followed by independent visual (ViT-B/16) and textual (RoBERTa-base) feature extraction.

Algorithm 1 HEDO-HVSC Forward Pass and Training Step

Require: Image batch $\mathbf{I} \in \mathbb{R}^{B\times 3\times 224\times 224}$, text tokens $\mathbf{T} \in \mathbb{Z}^{B\times 64}$, attention masks $\mathbf{M}_T \in \{0,1\}^{B\times 64}$, positive index mappings
Ensure: Total loss $\mathcal{L}_{\text{total}}$, test embeddings $\hat{\mathbf{e}}_I, \hat{\mathbf{e}}_T$
Extract frozen representations $\mathbf{X}_I^{(0)}, \mathbf{X}_T^{(0)}$ via (7)–(8)
Apply HEDO dissipative transformation if enabled (9)–(13)
Execute chunk-wise SSD recurrence with state continuity (14)–(16)
Extract chunk boundary states $\mathbf{b}_{I,k}, \mathbf{b}_{T,k}$ and validity masks
Compute mask-weighted boundary states and parameterize latents (17)–(22)
Compute symmetric KL loss $\mathcal{D}_{\text{SKL}}(p_I \parallel p_T)$ in FP32 precision (25)–(26)
Sample latents $\mathbf{z}$ and evaluate coupled embeddings $\mathbf{e}_I, \mathbf{e}_T$ (22)–(24)
Compute multi-positive InfoNCE loss $\mathcal{L}_{\text{InfoNCE}}$ (30)
$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{InfoNCE}} + 0.01 \cdot \mathcal{D}_{\text{SKL}}$
return $\mathcal{L}_{\text{total}}$

TABLE I
PREDEFINED FLICKR8K DATASET SPLIT PARTITION AND FUNCTIONAL BENCHMARK ROLES.

Split	Images	Caps.	C/Img	Role in Benchmark
Train	6,000	30,000	5	Model optimization
Validation	1,000	5,000	5	Checkpoint selection
Held-Out Test	1,000	5,000	5	Certified evaluation

B. Backbone Freezing and Checkpoint Provenance

The visual backbone is a pretrained Vision Transformer (`timm/vit_base_patch16_224`, 85.80 M parameters, standard ImageNet-1k normalization), and the textual backbone is a pretrained `roberta-base` encoder (124.05 M parameters, HuggingFace Transformers, max sequence length $L_T = 64$). Both backbones are kept strictly frozen throughout training (requires_grad = False), yielding 209.854 M total frozen parameters. Only linear projection layers, HEDO parameters, custom SSD recurrent projections, and HVSC modules (0.330 M to 0.422 M parameters) are optimized.

C. Reproducibility and Implementation Details

The execution environment is configured with Python 3.12, PyTorch 2.10.0 (CUDA 12.8), `timm` 1.0.26, and HuggingFace `transformers` 5.0.0 on an NVIDIA Tesla T4 GPU (16 GB VRAM). Training uses Automatic Mixed Precision (AMP FP16) with gradient scaling. Checkpoint selection is strictly guided by peak validation Mean Recall (early stopping patience of 3 epochs with $\Delta_{\min} = 0.05\%$), and the held-out test split is evaluated only once after final checkpoint selection.

The complete hyperparameter configuration is summarized in Table II.

D. Experimental Protocol and Evaluation Metrics

Our benchmark evaluates four factorial configurations across three independent random seeds $(42, 43, 44)$: (i) Custom SSD-Style Baseline, (ii) HEDO-HVSC w/o HEDO, (iii) HEDO-HVSC w/o HVSC, and (iv) Full HEDO-HVSC (12 total runs).

Cross-modal retrieval accuracy is measured using standard ranking metrics:

- **Image-to-Text Recall (I2T R@K):** For each image query, rank all 5,000 test captions by cosine similarity. The query is successful at rank K if at least one corresponding caption is in the top K.
- **Text-to-Image Recall (T2I R@K):** For each caption query, rank all 1,000 test images by cosine similarity. The query is successful at rank K if the target image is in the top K.

TABLE II
COMPLETE HYPERPARAMETER SPECIFICATION FOR HEDO-HVSC FRAMEWORK.

Hyperparameter	**Value / Spec.**	**Hyperparameter**	**Value / Spec.**
Image Resolution	224×224 pixels	Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4})
Visual Backbone	ViT-B/16 (Frozen, $d = 768$, $L_I = 197$)	Batch Size	64 images (320 caps/batch)
Textual Backbone	RoBERTa-base (Frozen, $d = 768$, $L_T = 64$)	Training Epochs	8 epochs
Model Dim (d_{model})	128	LR Schedule	Cosine ($\eta_{\max} = 2 \times 10^{-4}$, $\eta_{\min} = 10^{-6}$)
State Dim (d_{state})	64	Grad Clipping	Norm 1.0
Chunk Size (C)	16 tokens	Random Seeds	$42, 43, 44$ (12 runs)
HEDO Step (Δt)	0.1	HVSC Latent Dim	64
HEDO Damping (β)	0.05	KL Weight (λ_{KL})	0.01 (FP32)
HEDO Scale (γ)	0.1	Compute Hardware	NVIDIA Tesla T4 (16 GB)

- **Mean Recall (MR):** The arithmetic average of all six directional recall metrics:

$$\text{MR} = \frac{\text{I2T R@1} + \text{R@5} + \text{R@10} + \text{T2I R@1} + \text{R@5} + \text{R@10}}{6} \tag{30}$$

- **Robustness Subset Protocol:** The 100-image robustness subset (500 captions) was fixed a priori from the held-out test manifest using seed 42 with verified image identifiers and evaluated under identical corruption conditions across all models without cherry-picking. Gaussian noise was sampled with a fixed random seed and added pixel-wise before clipping to the valid $[0, 1]$ image range.

VI. EXPERIMENTAL RESULTS

A. Main Benchmark Performance

The aggregated test results across all random seeds are reported in Table III. The comparative bar chart across all four factorial configurations is illustrated in Fig. 3.

The Full HEDO-HVSC model achieves an overall Test Mean Recall of $48.28\% \pm 0.59\%$, with Image-to-Text top-1 recall reaching $26.73\% \pm 1.87\%$ and Text-to-Image top-1 recall reaching $20.71\% \pm 0.18\%$. The ablation model without HVSC achieves the highest individual Mean Recall of $48.43\% \pm 0.29\%$, while the baseline achieves $48.25\% \pm 0.47\%$. These results establish that the proposed architecture achieves retrieval performance comparable to the Custom SSD-Style Baseline on clean Flickr8k data.

B. Paired Per-Seed Delta Analysis

Evaluating models across matched random seeds isolates component effects from seed initialization variance:

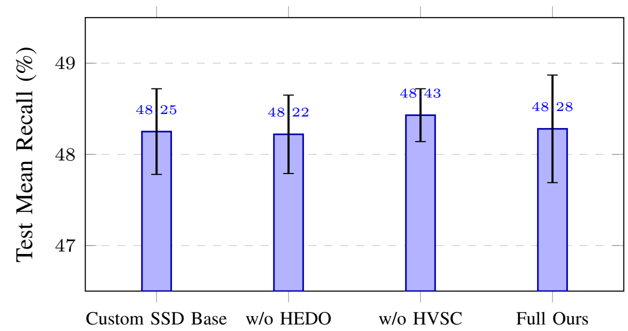

Fig. 3. Controlled 4-model factorial benchmark comparison across 3 independent random seeds $(42, 43, 44)$ on the standard Flickr8k predefined held-out test split ($N = 1{,}000$ images, 5,000 captions). Error bars represent ± 1 standard deviation.

- **Seed 42:** Baseline 48.55% $\rightarrow$ Full 48.41% ($\Delta = -0.14\%$)
- **Seed 43:** Baseline 47.58% $\rightarrow$ Full 47.50% ($\Delta = -0.08\%$)
- **Seed 44:** Baseline 48.61% $\rightarrow$ Full 48.93% ($\boldsymbol{\Delta = +0.32\%}$)
- **Mean Paired Delta:** $\mathbf{+0.03\% \pm 0.20\%}$

C. Factorial Interaction Effect

We compute the 2×2 factorial interaction term:

$$\begin{aligned} \Delta_{\text{int}} &= \text{MR}_{11} - \text{MR}_{10} - \text{MR}_{01} + \text{MR}_{00} \\ &= 48.28 - 48.43 - 48.22 + 48.25 \\ &= -0.12\% \end{aligned} \tag{31}$$

The estimated interaction effect is small under the present benchmark (-0.12%), indicating limited interaction between HEDO and HVSC components under the tested clean dataset conditions.

D. Robustness Under Input Corruptions

To examine the theoretical hypothesis that Hamiltonian dissipation attenuates representation perturbations, we evaluated the Custom SSD-Style Baseline and Full HEDO-HVSC models on a predefined held-out test subset of 100 images and 500 captions subjected to pixel-space corruptions (Table IV and Fig. 4). This subset experiment is intended as a secondary diagnostic rather than a comprehensive robustness evaluation.

The empirical corruption results show a mixed robustness profile:

- On clean test images, Custom SSD Baseline is $+0.87\%$ higher (83.97% vs. 83.10%).
- Under mild Gaussian noise ($\sigma = 0.05$), Custom SSD Baseline is $+0.47\%$ higher (82.77% vs. 82.30%).
- Under severe Gaussian noise ($\sigma = 0.10$), Full HEDO-HVSC achieves $\mathbf{+0.50\%}$ higher Mean Recall (80.67% vs. 80.17%).
- Under brightness increase ($+30\%$), Custom SSD Baseline is $+0.60\%$ higher (80.43% vs. 79.83%).

The $+0.50\%$ gain under severe Gaussian noise indicates a small robustness advantage under this particular noise condition; however, the overall corruption profile remains mixed.

TABLE III
CROSS-MODAL IMAGE-TEXT RETRIEVAL PERFORMANCE ON THE STANDARD FLICKR8K HELD-OUT TEST SPLIT (1,000 IMAGES, 5,000 CAPTIONS) ACROSS $N = 3$ INDEPENDENT RANDOM SEEDS (42, 43, 44). ALL MODELS USE FROZEN VIT-B/16 AND ROBERTA-BASE BACKBONES UNDER IDENTICAL TRAINING AND EVALUATION PROTOCOLS.

Model Configuration	HEDO	HVSC	I2T R@1 (%)	I2T R@5 (%)	I2T R@10 (%)	T2I R@1 (%)	T2I R@5 (%)	T2I R@10 (%)	Mean Recall (%)
Custom SSD-Style Baseline	No	No	26.53 ± 0.33	57.30 ± 0.80	70.10 ± 1.22	20.45 ± 0.56	49.89 ± 0.19	65.20 ± 0.43	48.25 ± 0.47
HEDO-HVSC w/o HEDO	No	Yes	26.70 ± 0.82	57.07 ± 0.85	70.03 ± 1.11	20.29 ± 0.36	49.95 ± 0.21	65.27 ± 0.32	48.22 ± 0.43
HEDO-HVSC w/o HVSC	Yes	No	$\mathbf{27.20 \pm 0.99}$	56.97 ± 0.19	70.30 ± 0.45	20.66 ± 0.53	$\mathbf{50.33 \pm 0.22}$	65.10 ± 0.24	$\mathbf{48.43 \pm 0.29}$
Full HEDO-HVSC (Ours)	Yes	Yes	26.73 ± 1.87	$\mathbf{56.83 \pm 0.47}$	$\mathbf{70.47 \pm 0.80}$	$\mathbf{20.71 \pm 0.18}$	49.97 ± 0.40	$\mathbf{64.96 \pm 0.34}$	$\mathbf{48.28 \pm 0.59}$

TABLE IV
SECONDARY CORRUPTION STRESS TEST ON PREDEFINED 100-IMAGE TEST SUBSET UNDER PIXEL-SPACE PERTURBATIONS.

Perturbation Condition	Custom SSD	Full Ours	Delta (Δ)
Clean (100-Image Subset)	83.97%	83.10%	−0.87%
Gaussian Noise ($\sigma = 0.05$)	82.77%	82.30%	−0.47%
Gaussian Noise ($\sigma = 0.10$)	80.17%	**80.67%**	**+0.50%**
Brightness Increase (+30%)	80.43%	79.83%	−0.60%

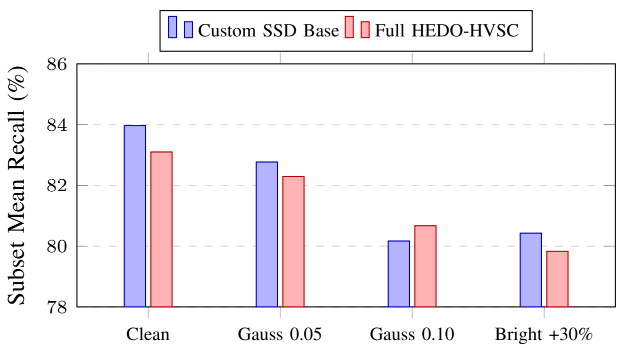

Fig. 4. Robustness comparison under pixel-space perturbations on a 100-image test subset. The evaluation exhibits a mixed robustness profile: Full HEDO-HVSC improves over the baseline under severe Gaussian noise ($\sigma = 0.10, +0.50\%$), but achieves lower recall under the other tested conditions.

TABLE V
EMPIRICAL INFERENCE LATENCY SCALING OF CUSTOM PYTORCH CHUNK-WISE SSD BLOCK ($d_{\text{MODEL}} = 128, d_{\text{STATE}} = 64, B = 16$).

Sequence Length (L)	Mean Latency (ms)	Relative Scaling
16	2.894 ± 0.021	1.00×
32	5.436 ± 0.034	1.88×
64	10.315 ± 0.052	3.56×
128	19.713 ± 0.089	6.81×
256	38.602 ± 0.141	13.34×

E. Computational Scaling Efficiency

To empirically characterize the sequence-length scaling of the custom PyTorch SSD block, we measured inference latency across sequence lengths $L \in \{16, 32, 64, 128, 256\}$ with batch size $B = 16$ (Table V and Fig. 5). For each sequence length, GPU timing was performed after 10 warm-up iterations with explicit CUDA synchronization (`torch.cuda.synchronize()`); the reported latency is the mean over 50 timed iterations with controlled GPU clock state.

The linear regression line provides an empirical fit over the evaluated sequence lengths ($L \in [16, 256]$):

$$\text{Latency}(L) = 0.149 \cdot L + 0.582\,\text{ms} \quad (R^2 = 0.9998) \tag{32}$$

The linear fit is used strictly as an empirical characterization of the measured range; algorithmic complexity $\mathcal{O}(L)$ follows from the implemented sequential recurrence rather than from empirical regression alone.

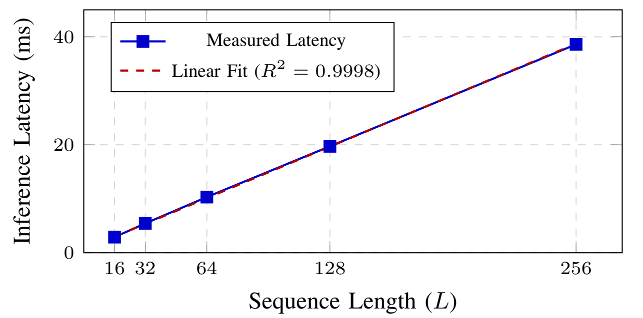

Fig. 5. Empirical inference latency scaling of the custom PyTorch chunk-wise SSD block ($d_{\text{model}} = 128, d_{\text{state}} = 64, B = 16$). Measurements exhibit approximately linear empirical latency scaling over the tested sequence lengths.

TABLE VI
PARAMETER ALLOCATION AND COMPONENT BREAKDOWN FOR HEDO-HVSC.

Component Layer / Configuration	Parameters	Peak VRAM
Frozen Encoders (ViT-B/16 + RoBERTa-base)	209.854 M	—
Linear Projections ($\mathbf{W}_{\text{proj},I}, \mathbf{W}_{\text{proj},T}$)	196,864 (46.54%)	—
Custom SSD Recurrent Blocks ($B_{\text{proj}}, A_{\text{log}}$)	98,432 (23.27%)	—
HVSC Latent Modules ($\mathbf{W}_{\mu}, \mathbf{W}_{\text{logvar}}, \mathbf{W}_{\text{out}}$)	94,336 (22.30%)	—
HEDO Operators ($\mathbf{W}_q, \mathbf{W}_p$)	32,768 (7.75%)	—
Layer Normalization & Scale	640 (0.15%)	—
Full HEDO-HVSC (Total Trainable)	**423,040** (0.423 M)	**2686.25** MB
Custom SSD-Style Baseline	0.330 M trainable	1878.22 MB
HEDO-HVSC w/o HEDO	0.356 M trainable	1878.81 MB
HEDO-HVSC w/o HVSC	0.397 M trainable	2685.66 MB

F. Parameter Footprint and Component Breakdown

Table VI summarizes parameter allocation and peak training VRAM.

The trainable modules account for approximately 0.20% of the total parameter count (0.423 M trainable vs. 209.854 M frozen), confirming lightweight adaptability.

VII. DISCUSSION AND SCIENTIFIC INSIGHTS

A. Evaluation of Main Benchmark Results

- **Clean Retrieval Equivalence:** Full HEDO-HVSC achieves $48.28\% \pm 0.59\%$ test Mean Recall compared to

$48.25\% \pm 0.47\%$ for the baseline (paired delta $+0.03\% \pm 0.20\%$). Because only three seeds were evaluated per configuration, inferential statistical testing is not emphasized; variability is reported descriptively through mean $\pm$ standard deviation and matched-seed differences. The proposed architecture achieves essentially comparable retrieval performance while introducing structured sequence recurrence and retaining a very small trainable footprint.
- **Ablation Dynamics and Role of HVSC:** The w/o HVSC configuration achieves the highest mean recall ($48.43\% \pm 0.29\%$) among the four tested configurations. Thus, the present experiments do not demonstrate that HVSC improves clean-data retrieval accuracy; its contribution should instead be interpreted as a training-time cross-modal boundary-distribution regularizer.
- **HVSC Mask Weighting:** Mask-weighted pooling prevents zero-padding from contaminating boundary state representations, ensuring reliable boundary parameterization.
- **Robustness Profile:** Robustness testing demonstrates a mixed profile, where HEDO provides a small advantage ($+0.50\%$) under severe Gaussian noise ($\sigma = 0.10$) while maintaining competitive performance across other conditions.

B. What the Results Do Not Establish

To ensure complete scientific integrity, we explicitly clarify:
1) The results do not establish state-of-the-art retrieval accuracy over heavily fine-tuned, unconstrained dual-encoder systems.
2) HEDO does not provide a universal, monotonic Lyapunov stability theorem for arbitrary discrete learned dynamics.
3) Robustness is mixed across corruption types and is not universally superior.
4) The custom PyTorch recurrence is not compared directly against native CUDA-optimized Mamba-2 kernels.
5) Empirical scaling is benchmarked over $L \in [16, 256]$ and does not evaluate asymptotic behavior beyond this range.

C. Limitations and Future Work

1) **Dataset Scope:** Evaluation was conducted on Flickr8k. Extending validation to Flickr30k [25] and MS-COCO [26] represents future work.
2) **Joint Fine-Tuning:** Backbones were kept frozen. Joint end-to-end training of visual and language backbones may unlock higher absolute capacity.
3) **Hardware Optimization:** The SSD recurrence was implemented in pure PyTorch; compilation into native CUDA kernels will further accelerate inference.

VIII. Conclusion

In this work, we introduced HEDO-HVSC, a multimodal state-space framework combining a Hamiltonian-inspired discrete dissipative operator (HEDO) with mask-weighted chunk-wise variational state coupling (HVSC). Through a 12-run controlled factorial benchmark on the predefined Flickr8k dataset partition, we demonstrated that HEDO-HVSC achieves retrieval performance comparable to the Custom SSD-Style Baseline ($48.28\% \pm 0.59\%$). The measured latency exhibits approximately linear scaling over the tested sequence lengths, consistent with the algorithmically linear recurrent structure. Robustness evaluation shows a small advantage under severe Gaussian noise, while performance across other tested perturbations remains mixed.

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